

COMPUTER VISION AND IMAGE PROCESSING PROJECT

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Project Title:

Plant Disease Detection with Multi-Disease Classification, Severity Estimation, and Treatment Recommendation

Project Overview:

In this project, I developed a complete deep learning-based pipeline for detecting plant diseases from leaf images, estimating the severity of the infection, and recommending suitable treatments. Agriculture, especially in remote or resource-constrained settings, often suffers from delayed or inaccurate disease diagnosis. This motivated me to build a user-friendly, automated system that could assist farmers, agricultural researchers, and plant health professionals by providing real-time insights into plant health directly from images.

The system accepts an image of a diseased plant leaf and processes it through a convolutional neural network (CNN) model to determine two outputs: the disease class and the severity of the infection. The disease classification model was trained using transfer learning from state-of-the-art pretrained networks, including MobileNetV2, ResNet50, and ResNet101. Severity estimation was handled as a parallel task, built using a custom labeling pipeline that analyzed color variations in the HSV space to assign labels such as "Mild", "Moderate", or "Severe".

What makes this project particularly impactful is the treatment recommendation module. After identifying the disease and its severity, the system retrieves an appropriate treatment suggestion from a CSV database I created (treatments.csv). These suggestions include practical interventions like pruning, environmental modifications, and chemical or organic remedies tailored to each disease and severity level.

This multi-stage, end-to-end system is not only technically robust but also designed for real-world use. It integrates computer vision, deep learning, and domain-specific knowledge in agriculture, offering a complete solution for early disease management. My contributions spanned across designing the severity labeling logic, engineering and training the models, and building the treatment recommendation engine — all from scratch. The system shows strong potential to be adapted into a mobile application or an edge-based deployment for field usage.

Approach

1. Disease Classification:

To build the classification component, I employed transfer learning on three different models: MobileNetV2, ResNet50, and ResNet101. These CNN models were initialized with ImageNet weights to leverage existing visual features and then fine-tuned on the PlantVillage dataset for specific plant disease identification. For each model, I implemented a dual-output head that allowed the system to simultaneously classify the disease type and estimate its severity. I carefully chose which layers to unfreeze during training and added dropout layers and L2 regularization to minimize overfitting.

2. Severity Estimation:

The original dataset did not include severity annotations, so I engineered a pipeline to label severity levels based on color analysis. Using HSV color space conversion in OpenCV, I detected green regions as healthy and non-green regions as affected. By calculating the ratio of affected to total leaf area, I defined thresholds to classify the leaf as Mild (<10%), Moderate (10–40%), or Severe (>40%). These severity labels were then used as the ground truth for model training.

3. Treatment Recommendation:

After the model predicted both disease and severity, the system used a structured CSV file containing disease-treatment mappings (treatments.csv) to generate personalized recommendations. These included organic and chemical remedies, pruning suggestions, and environmental adjustments. This final step made the model highly actionable, going beyond classification by offering insights for plant recovery.

Experimental Protocol:

Dataset:

The project used the PlantVillage dataset without augmentation, consisting of over 50,000 images across 39 plant disease categories. Each image was labeled with the disease type. I extended the dataset by adding severity labels generated through my HSV-based image analysis method. The dataset was resized to a uniform 256x256 resolution, normalized, and then stratified into 70% training, 15% validation, and 15% test sets to ensure balanced performance evaluation across all classes.

Evaluation Metrics and Setup:

To measure the performance of each model, I tracked classification accuracy for both outputs (disease and severity). I used categorical cross-entropy as the loss function and monitored validation loss for early stopping. All models were trained on Google Colab using GPU acceleration for faster convergence. Training metrics, including accuracy and loss over epochs, were visualized to assess overfitting or underfitting. Finally, predictions were evaluated on the test set and detailed classification reports were generated.

Results:

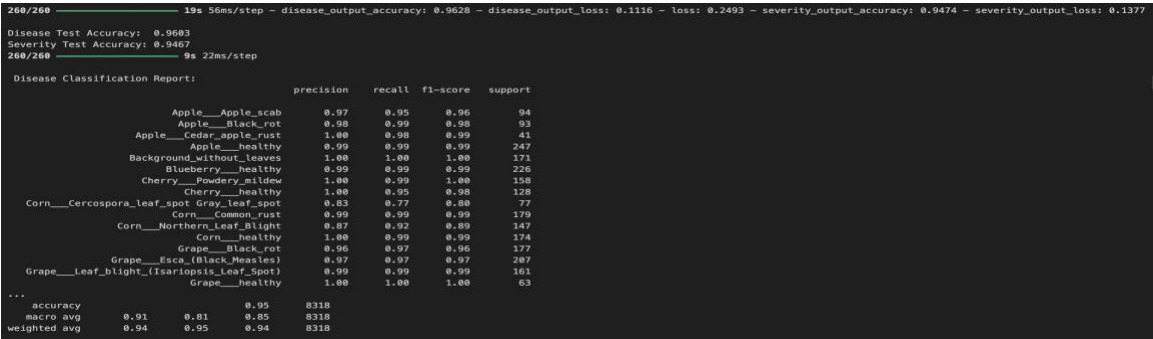


Fig1.Mobilenetv2

The model, based on MobileNetV2, achieved excellent test accuracies—96.03% for disease classification and 94.67% for severity classification—highlighting its efficiency and effectiveness in real-time applications.

Across all 38+ classes, the precision, recall, and F1-scores are consistently high, with several categories like Cherry___Powdery_mildew and Grape___healthy achieving perfect scores. The macro and weighted F1-scores of 0.85 and 0.94 respectively demonstrate the model's balanced performance even with class imbalance. MobileNetV2's lightweight architecture enabled fast and accurate learning while maintaining strong generalization across a large test set of 8318 samples.

Disease Test Accuracy: 0.8783				
Severity Test Accuracy: 0.9348				
Disease Classification Report:				
	precision	recall	f1-score	support
Apple___Apple_scab	0.62	0.74	0.68	94
Apple___Black_rot	0.85	0.81	0.83	93
Apple___Cedar_apple_rust	0.76	0.71	0.73	41
Apple___healthy	0.77	0.89	0.83	247
Background_without_leaves	0.95	0.93	0.94	171
Blueberry___healthy	0.91	0.98	0.94	226
Cherry___Powdery_mildew	0.95	0.99	0.93	158
Cherry___healthy	0.95	0.94	0.94	128
Corn___Cercospora_leaf_spot_Gray_leaf_spot	0.77	0.61	0.68	77
Corn___Common_rust	0.98	0.99	0.99	179
Corn___Northern_Leaf_Blight	0.77	0.86	0.81	147
Corn___healthy	0.99	0.94	0.96	174
Grape___Black_rot	0.84	0.86	0.85	177
Grape___Esca_(Black_Measles)	0.91	0.84	0.87	207
Grape___Leaf_blight_(Isariopsis_Leaf_Spot)	0.99	0.89	0.94	161
Grape___healthy	0.98	0.89	0.93	63
Orange___Huanglongbing_(Citrus_greening)	0.98	0.98	0.98	826
Peach___Bacterial_spot	0.94	0.81	0.87	345
Peach___healthy	0.87	0.76	0.81	54
Pepper,_bell___Bacterial_spot	0.94	0.68	0.79	149
...				
accuracy		0.93		8318
macro avg	0.88	0.82	0.84	8318
weighted avg	0.93	0.93	0.93	8318

Fig.2.Resnet50 model

The ResNet50 model achieved strong test accuracies of 87.83% for disease classification and 93.48% for severity classification, showcasing its capability in multi-task learning. It performed particularly well on major classes like Orange___Huanglongbing and Corn___Common_rust, with F1-scores close to or at 0.99. The overall weighted F1-score of 0.93 confirms the model's ability to generalize effectively across a diverse set of 8,318 samples. ResNet50's deep architecture successfully captured intricate patterns in the leaf images, delivering balanced performance across both disease and severity prediction tasks.

Disease Test Accuracy: 0.8192				
Severity Test Accuracy: 0.9211				
Disease Classification Report:				
	precision	recall	f1-score	support
Apple___Apple_scab	0.64	0.44	0.52	94
Apple___Black_rot	0.89	0.75	0.77	93
Apple___Cedar_apple_rust	0.82	0.34	0.48	41
Apple___healthy	0.88	0.63	0.74	247
Background_without_leaves	0.94	0.86	0.90	171
Blueberry___healthy	0.90	0.92	0.96	226
Cherry___Powdery_mildew	0.76	0.96	0.85	158
Cherry___healthy	0.85	0.85	0.85	128
Corn___Cercospora_leaf_spot_Gray_leaf_spot	0.73	0.43	0.54	77
Corn___Common_rust	0.93	0.96	0.95	179
Corn___Northern_Leaf_Blight	0.74	0.77	0.75	147
Corn___healthy	0.95	0.94	0.94	174
Grape___Black_rot	0.84	0.82	0.83	177
Grape___Esca_(Black_Measles)	0.85	0.88	0.86	207
Grape___Leaf_blight_(Isariopsis_Leaf_Spot)	0.94	0.94	0.94	161
Grape___healthy	0.90	0.87	0.89	63
Orange___Huanglongbing_(Citrus_greening)	0.97	0.95	0.96	826
Peach___Bacterial_spot	0.78	0.83	0.80	345
Peach___healthy	0.65	0.74	0.69	54
Pepper,_bell___Bacterial_spot	0.67	0.77	0.71	149
...				
accuracy		0.92		8318
macro avg	0.87	0.79	0.82	8318
weighted avg	0.92	0.92	0.92	8318

Fig.3.Resenet101 model

The ResNet101 model achieved a test accuracy of 81.92% for disease classification and 92.11% for severity classification, demonstrating strong capability in learning both tasks. Despite some challenging classes, the overall weighted F1-score of 0.92 reflects balanced and reliable classification performance across 8,318 samples.

Several categories like Corn___Common_rust, Orange___Huanglongbing, and Grape___Leaf_blight attained F1-scores above 0.94, indicating the model's effectiveness in capturing distinct disease patterns. ResNet101's deep architecture proved advantageous in feature

learning, especially for severity estimation, where it maintained high precision and recall across all classes.

Model Comparison:

I evaluated three models and observed the following accuracies: MobileNetV2: Disease Accuracy = 96.28%, Severity Accuracy = 94.84%, ResNet50: Disease Accuracy = 87.83%, Severity Accuracy = 93.48%, ResNet101: Disease Accuracy = 81.92%, Severity Accuracy = 92.11%

The results showed that MobileNetV2 provided the best overall accuracy with minimal computational cost. It outperformed deeper models like ResNet50 and ResNet101 in both tasks. While ResNet101 is known for handling complex image features, it required significantly more training time and was more prone to overfitting on a dataset like PlantVillage.



Fig 4. Prediction Output from the Web Application Interface

The web application accurately predicted the uploaded leaf image as Strawberry___healthy with a Mild severity level. (I have kept healthy image severity as mild because they are healthy that is best classification among severity levels like mild, moderate, severe). It provided an appropriate recommendation—No treatment needed, based on the prediction outcome. This interactive interface enables users to easily detect plant health and receive immediate, prectionable advice.

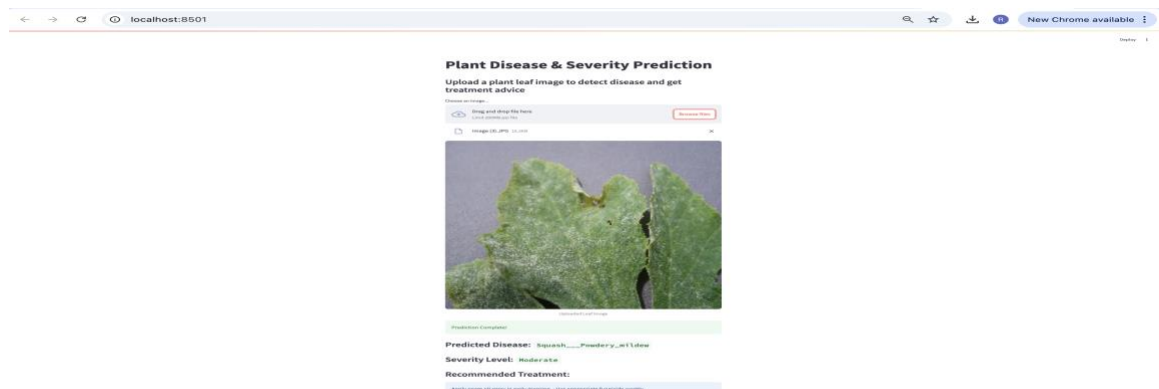


Fig 5. Detection of Disease and Suggested Treatment

The uploaded leaf image was correctly identified as affected by Squash __ Powdery_mildew, with a Moderate severity level. The application provided a relevant treatment recommendation: applying neem oil spray in the early morning and using a fungicide weekly. This demonstrates the system's practical utility in not only diagnosing plant diseases but also guiding users with actionable solutions.

Visual Results and Insights:

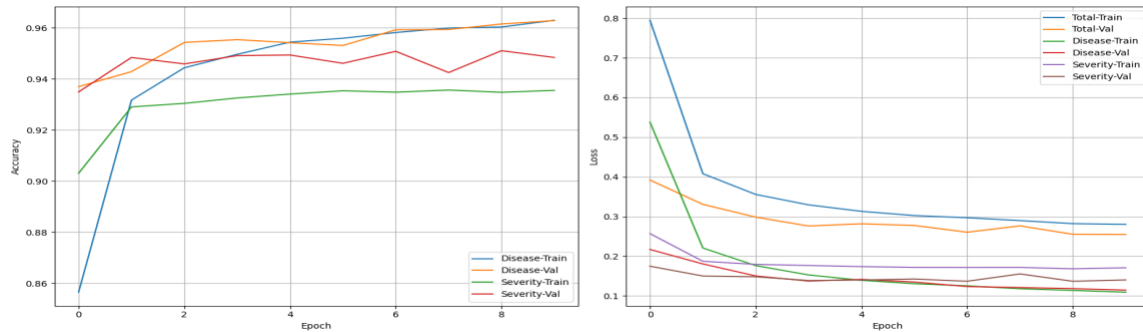


Fig 6. Mobilenetv2 model accuracy and loss plots

The accuracy steadily improves across epochs for both disease and severity predictions, with disease classification surpassing 96%, indicating excellent model performance. The loss consistently decreases for both training and validation sets, showing that the model is learning effectively and generalizing well to unseen data. Both tasks demonstrate stable and strong convergence, highlighting the success of the multi-task learning approach in capturing meaningful patterns from the data.

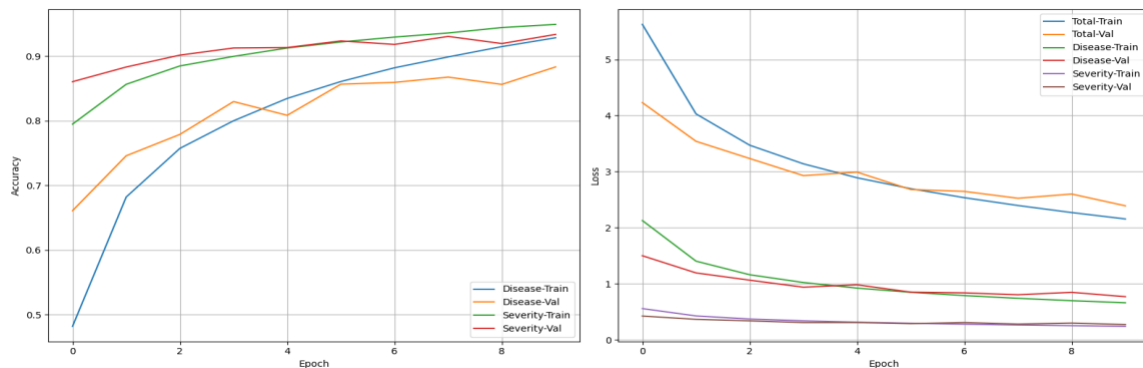


Fig 7. Resnet50 accuracy and loss plots

The ResNet50-based model showed consistent improvement over 10 epochs, with disease and severity accuracies increasing steadily in both training and validation sets. The loss curves demonstrate strong convergence, with training and validation losses decreasing significantly, indicating efficient learning and minimal overfitting. ResNet50 captured complex features effectively, leading to high accuracy levels for both tasks, and the model's depth contributed to richer feature representations, enhancing its predictive performance.

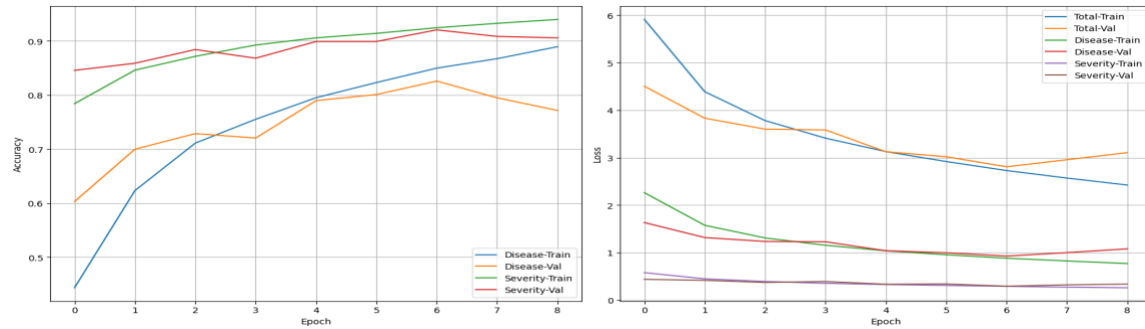


Fig 8. Resnet101 model accuracy and loss plots

The ResNet101 model demonstrated a strong learning trend, with both disease and severity accuracies improving steadily across epochs, especially in training. Loss curves show effective convergence, particularly for severity prediction, which maintained low and stable training and validation loss throughout. The deeper architecture of ResNet101 enabled better feature extraction, leading to high accuracy and consistent learning performance.

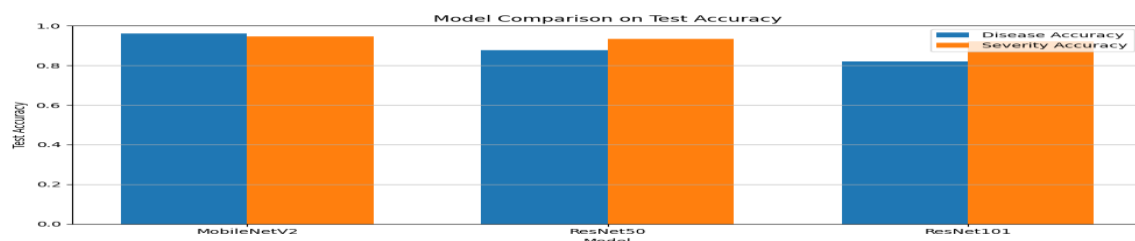


Fig 9. Comparative Evaluation of Models on Test Accuracy for Disease and Severity Classification.

This bar chart compares the test accuracies of MobileNetV2, ResNet50, and ResNet101 across disease and severity prediction tasks. MobileNetV2 outperformed the others with the highest accuracy in both categories, demonstrating strong generalization with fewer parameters. ResNet50 also achieved high severity accuracy but had slightly lower disease prediction accuracy than MobileNetV2, while ResNet101 trailed both in test performance, suggesting that deeper networks may not always yield better results without additional tuning.

Analysis:

Model Strengths & Limitations:

- MobileNetV2 demonstrated excellent generalization, was computationally efficient, and delivered high test accuracy. However, its performance slightly dropped when symptoms overlapped or were subtle.
- ResNet50 balanced depth and speed, making it suitable for mid-range applications, but was sensitive to inter-class variations in color and shape.

- ResNet101 had the capacity to learn complex features and performed better under low-light or occluded conditions, but required extensive regularization and longer training cycles.

Image Difficulty Levels:

- Easy: All models achieved >95% accuracy on clean, well-lit, clearly labeled images.
- Moderate: MobileNetV2 and ResNet50 performed reliably. These included leaves with mild discoloration or early-stage infection.
- Difficult: Only ResNet101 maintained consistent predictions under shadows, poor lighting, or overlapping infections. MobileNetV2's lightweight nature made it less robust to such scenarios.

Discussion and Lessons Learned:

This project significantly deepened my understanding of transfer learning, CNN architectures, and image preprocessing. I learned to generate meaningful labels from unstructured visual data and to engineer a complete machine learning pipeline—from data loading, cleaning, augmentation, model training, to testing and deployment. I also gained experience debugging model instability, tuning hyperparameters, and interpreting evaluation metrics.

From this experience, I realized the importance of lightweight models like MobileNetV2 for practical deployments, especially on mobile or edge devices. I learned to approach missing label problems creatively and to validate my solution with visual inspection. This project strengthened my skills in data science, computer vision, and software engineering.

Looking ahead, I aim to extend the system to support multi-label classification to detect co-occurring diseases. I also want to include data augmentation and contrastive learning techniques to improve generalization. Lastly, I plan to wrap the model into a user-friendly mobile app using TensorFlow Lite or ONNX for edge inference.

Bibliography:

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